

2.3. Markov Dynamics. If $(X, \mathcal{B})$ is a measurable space and U is a topological space, a *stochastic kernel* P from U to X is a function $P: U \times \mathcal{B} \rightarrow \mathbb{R}_+$ with the property that $u \mapsto P(u, B)$ is Borel measurable for each $B \in \mathcal{B}$ and $B \mapsto P(u, B)$ is a probability on $(X, \mathcal{B})$ for all $u \in U$. When $U = X$, we say that P is a stochastic kernel on X . Given a stochastic kernel P from U to X , we set

$$(Ph)(u) := \int h(x')P(u, dx') \quad (h \in bmX, u \in U).$$

When X is a topological space we use the following standard terminology: We call P *Feller* if Ph is continuous on U whenever $h \in bmX$ is continuous on X . We call P *strong Feller* if Ph is continuous on U whenever $h \in bmX$.

3. ABSTRACT DYNAMIC PROGRAMS

This section is a brief review of the dynamic programming framework from [Sargent and Stachurski \(2025\)](#). We repeat the key definitions to make the paper partly self-contained.

3.1. Definition and Properties. An *abstract dynamic program* (ADP) is a pair $(V, \mathbb{T})$, where

- (i) $V = (V, \preceq)$ is a poset and
- (ii) $\mathbb{T} = \{T_\sigma : \sigma \in \Sigma\}$ is a nonempty family of order preserving self-maps on V .

We call V the *value space*. Each operator T_σ in $\mathbb{T}$ is called a *policy operator*. Σ is an arbitrary index set and elements of Σ are called *policies*. In applications, we impose conditions under which each T_σ has a unique fixed point. In these settings, the significance of each policy operator T_σ is that its fixed point, denoted below by v_σ , represents the lifetime value of following policy σ .

Let $(V, \mathbb{T})$ be an ADP with policy set Σ . Given $v \in V$, a policy $\sigma \in \Sigma$ is called *v-greedy* if $T_\tau v \preceq T_\sigma v$ for all $\tau \in \Sigma$. An element $v \in V$ is said to satisfy the *Bellman equation* if

$$v := \bigvee_{\sigma} T_\sigma v \quad (v \in V). \tag{1}$$

In (1), the supremum is taken over all $\sigma \in \Sigma$. The *Bellman operator* generated by $(V, \mathbb{T})$ is defined via

$$Tv := \bigvee_{\sigma} T_\sigma v \quad \text{whenever the supremum exists.}$$